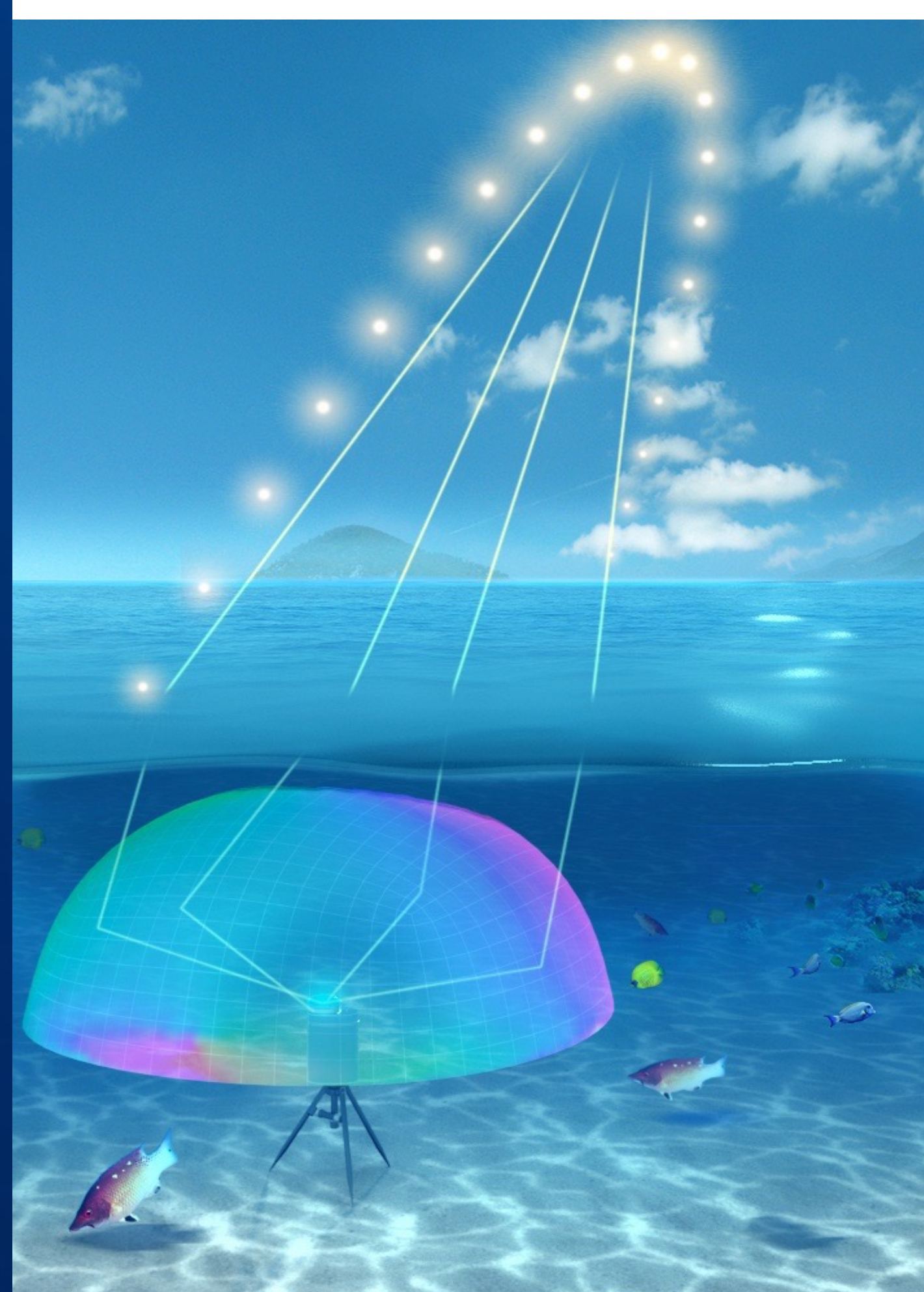


1. Motivation

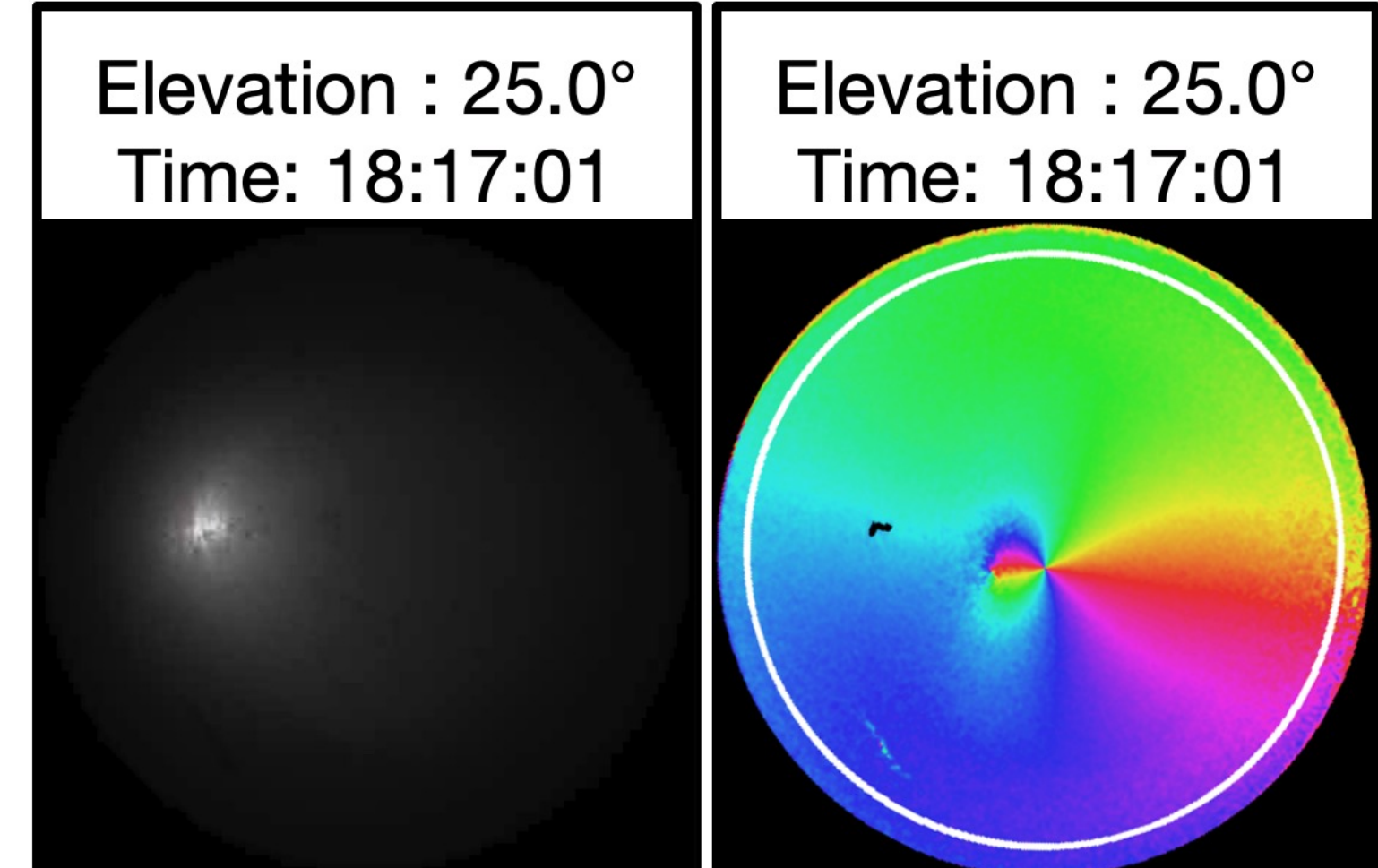
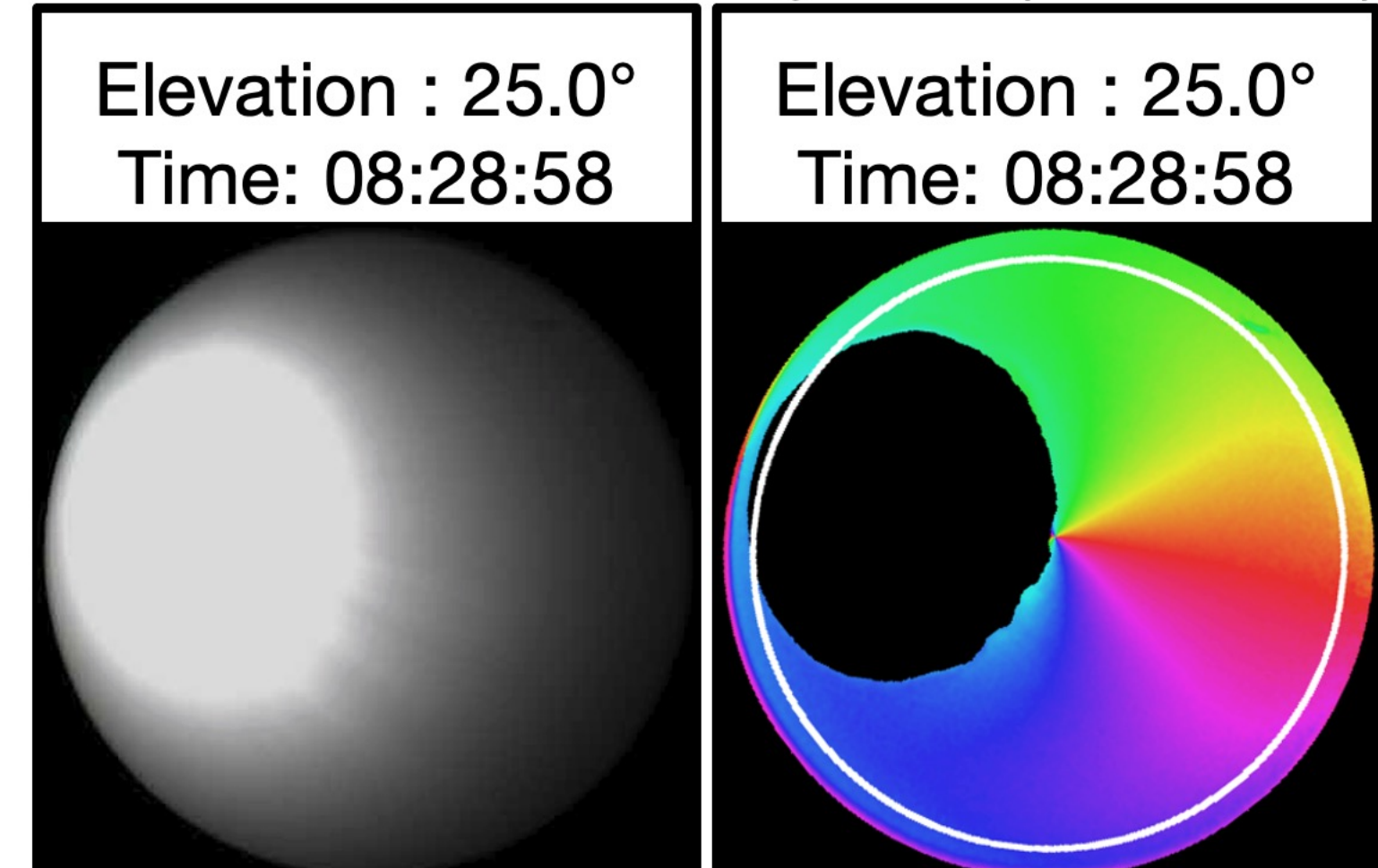
- No GPS underwater

Polarization time-lapse imagery
↓
Solar-elevation trajectory
↓
Geolocation

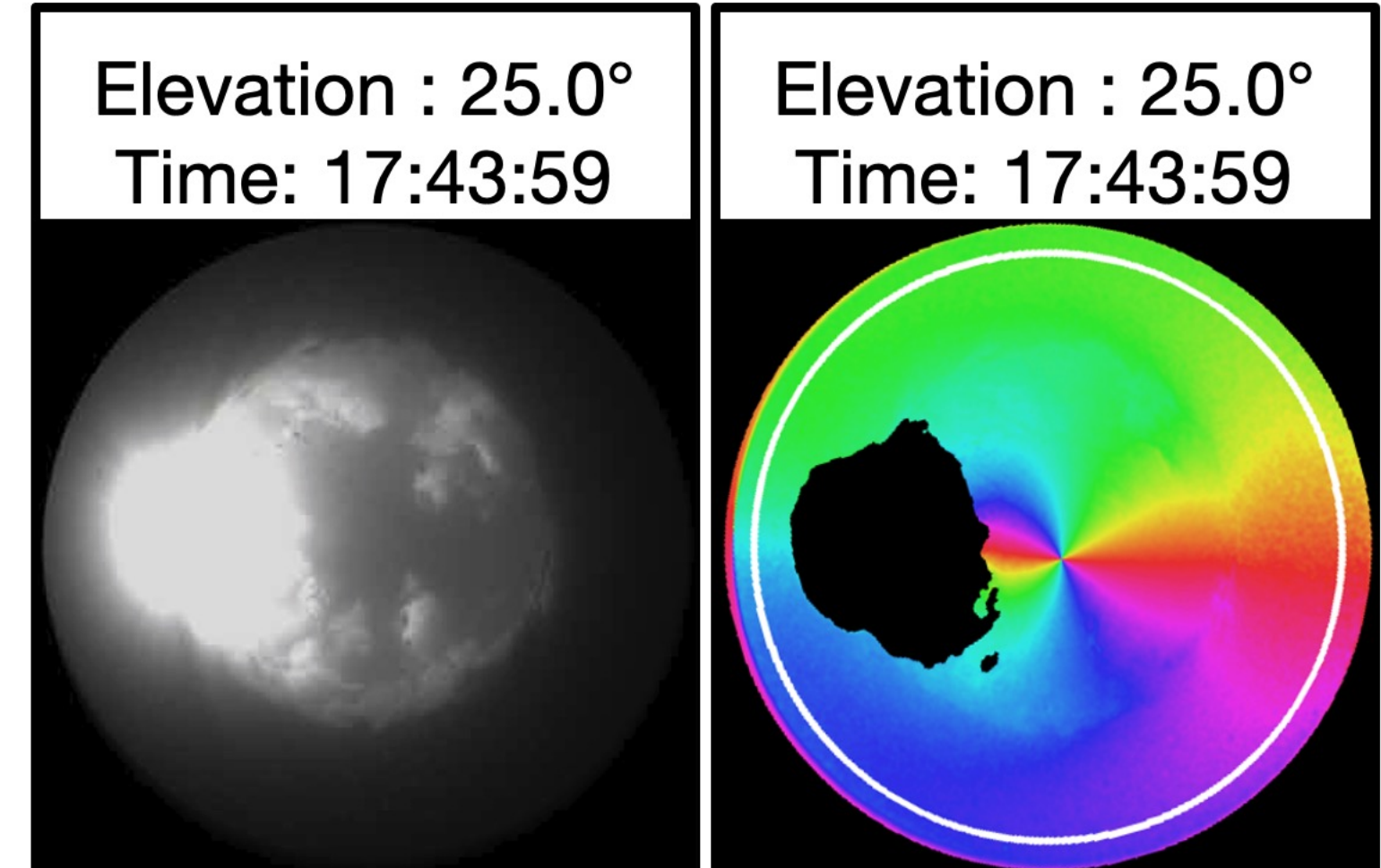
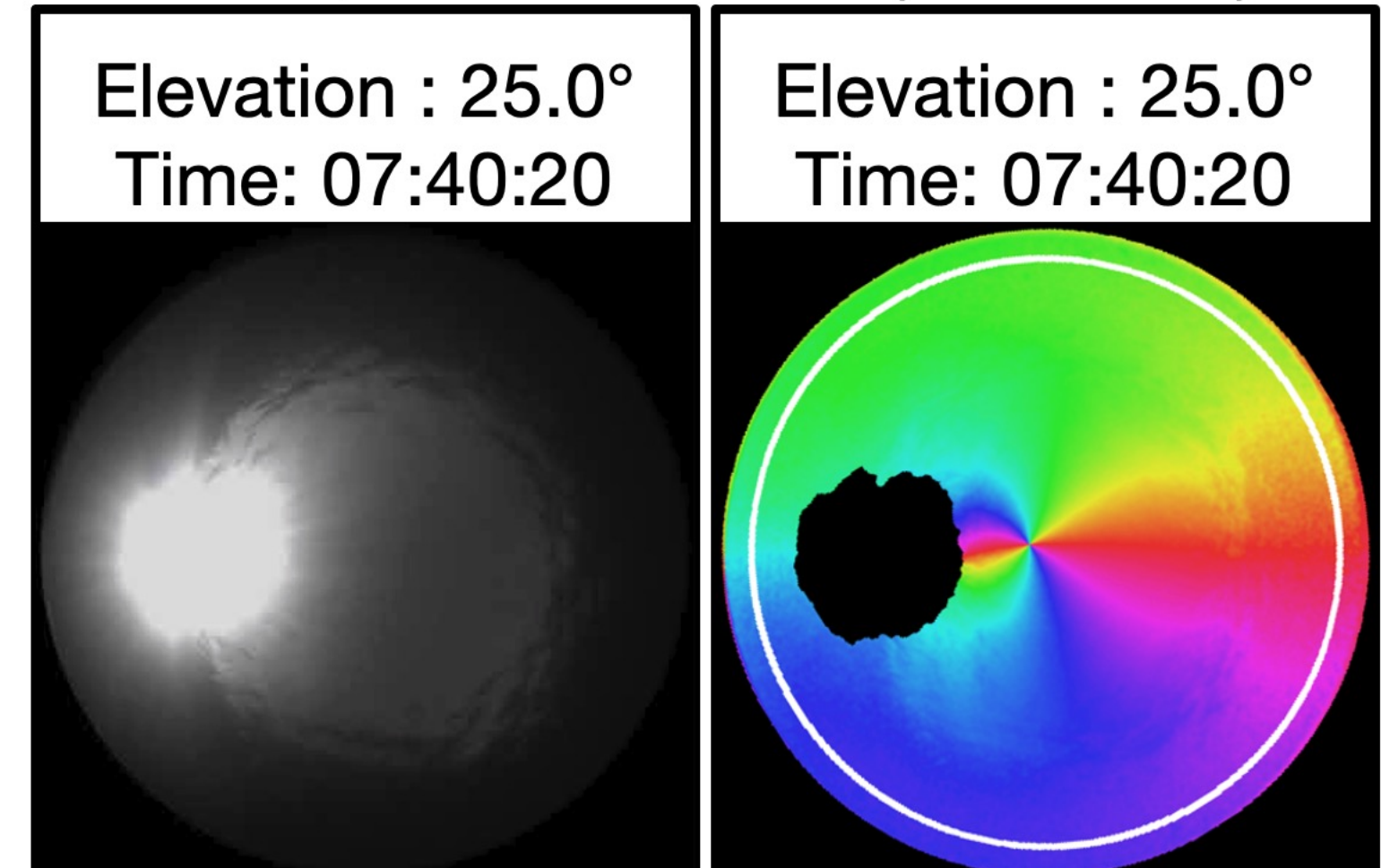


2. Angle of Polarization (AoP)

Rio Ceballo, Cordoba, Argentina (01/06/2023)

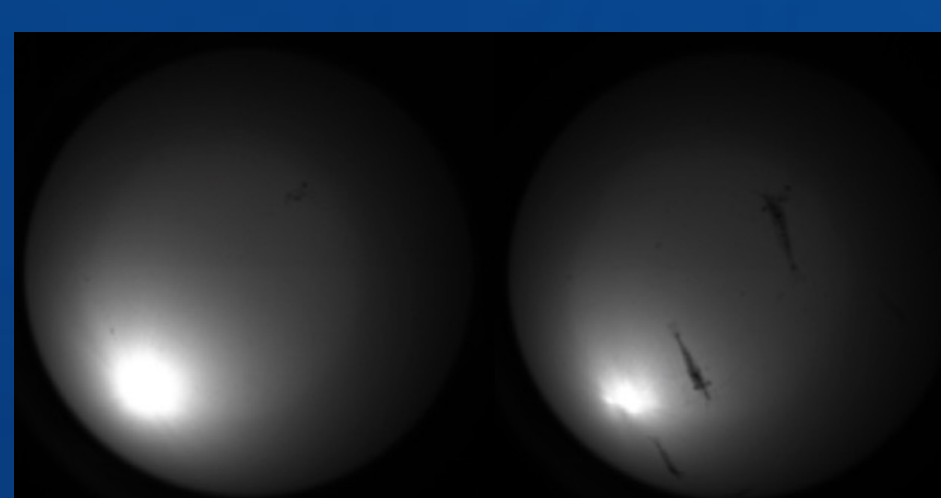


Ohrid, North Macedonia (07/12/2022)

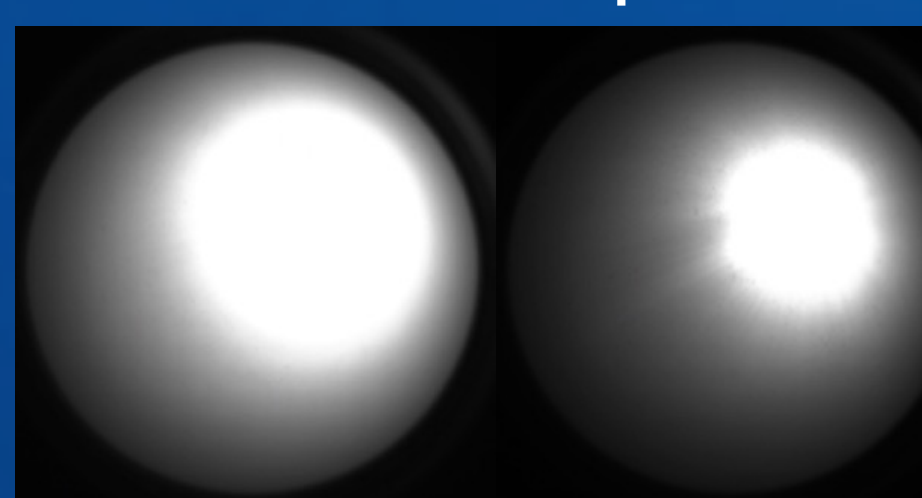


Why single-frame prediction is challenging?

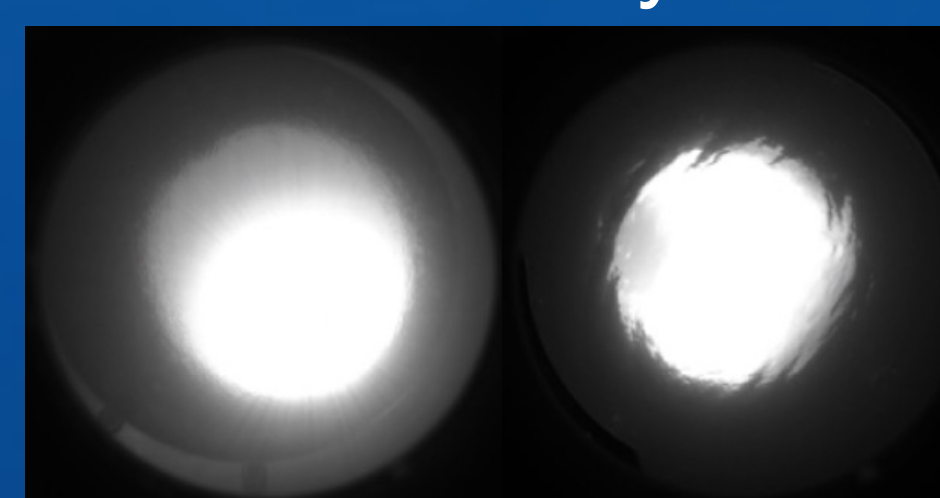
Marine life occlusion



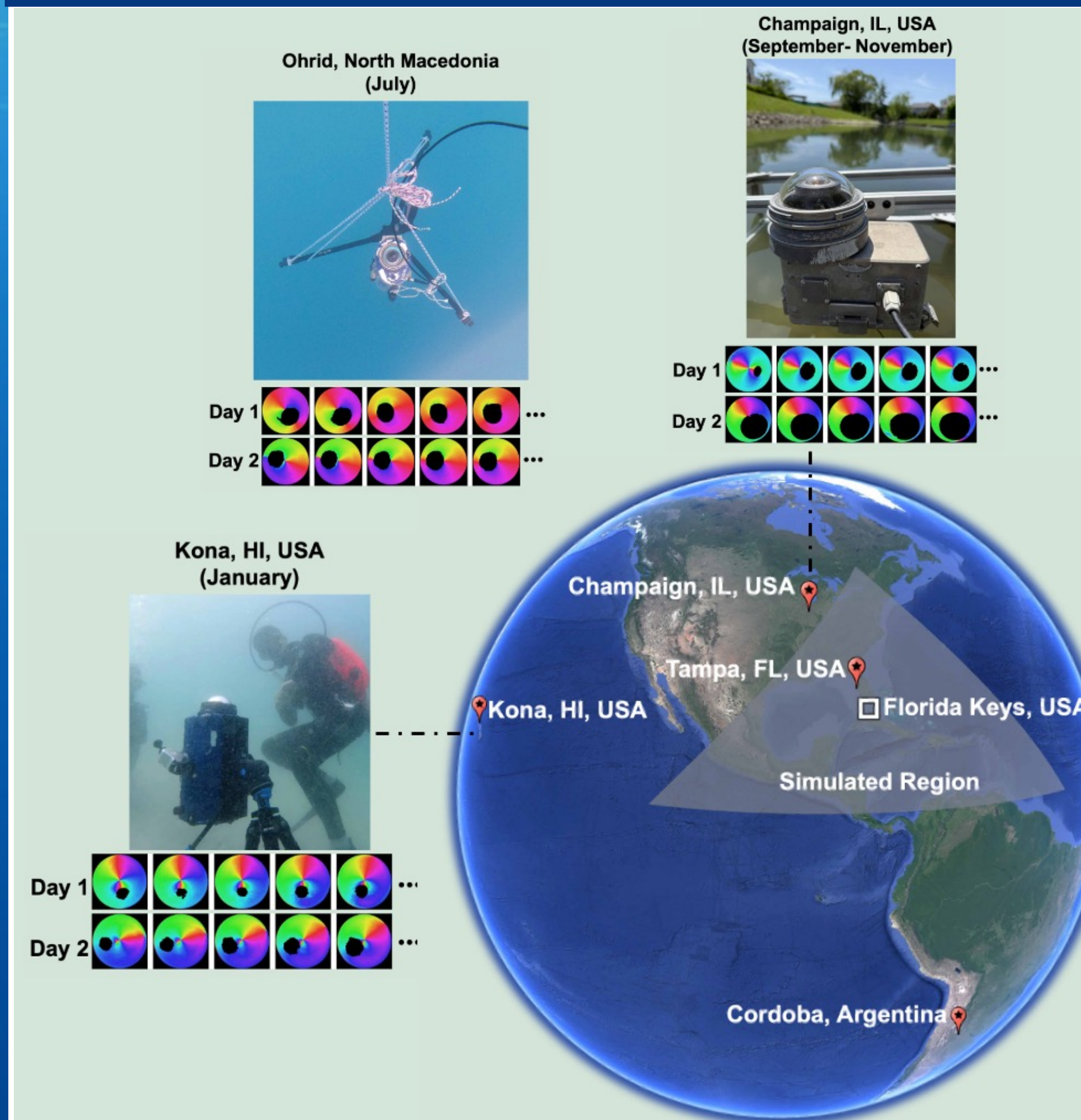
Variable exposure



Turbidity

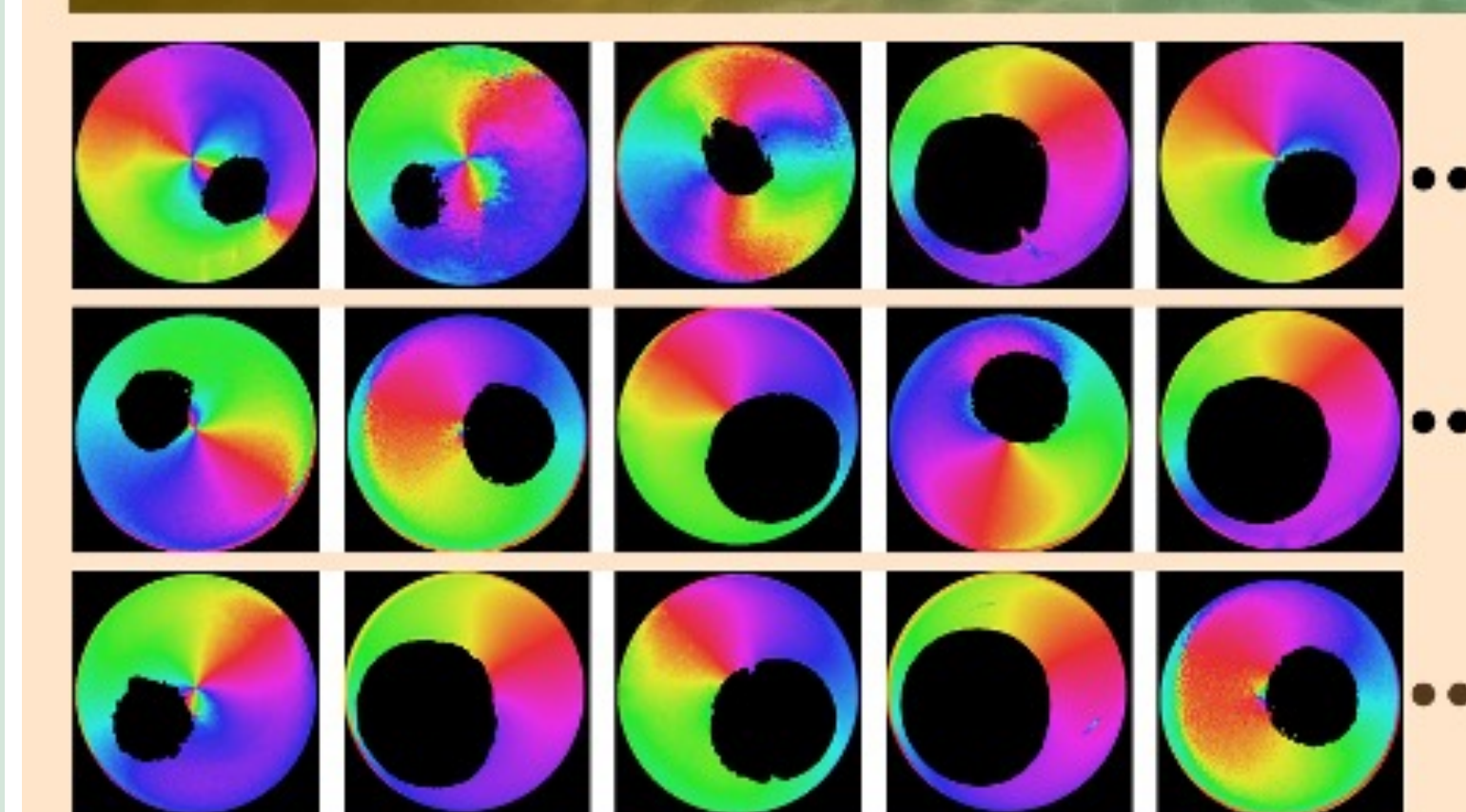


3. Physics-Guided Global Simulation of Training Sequences



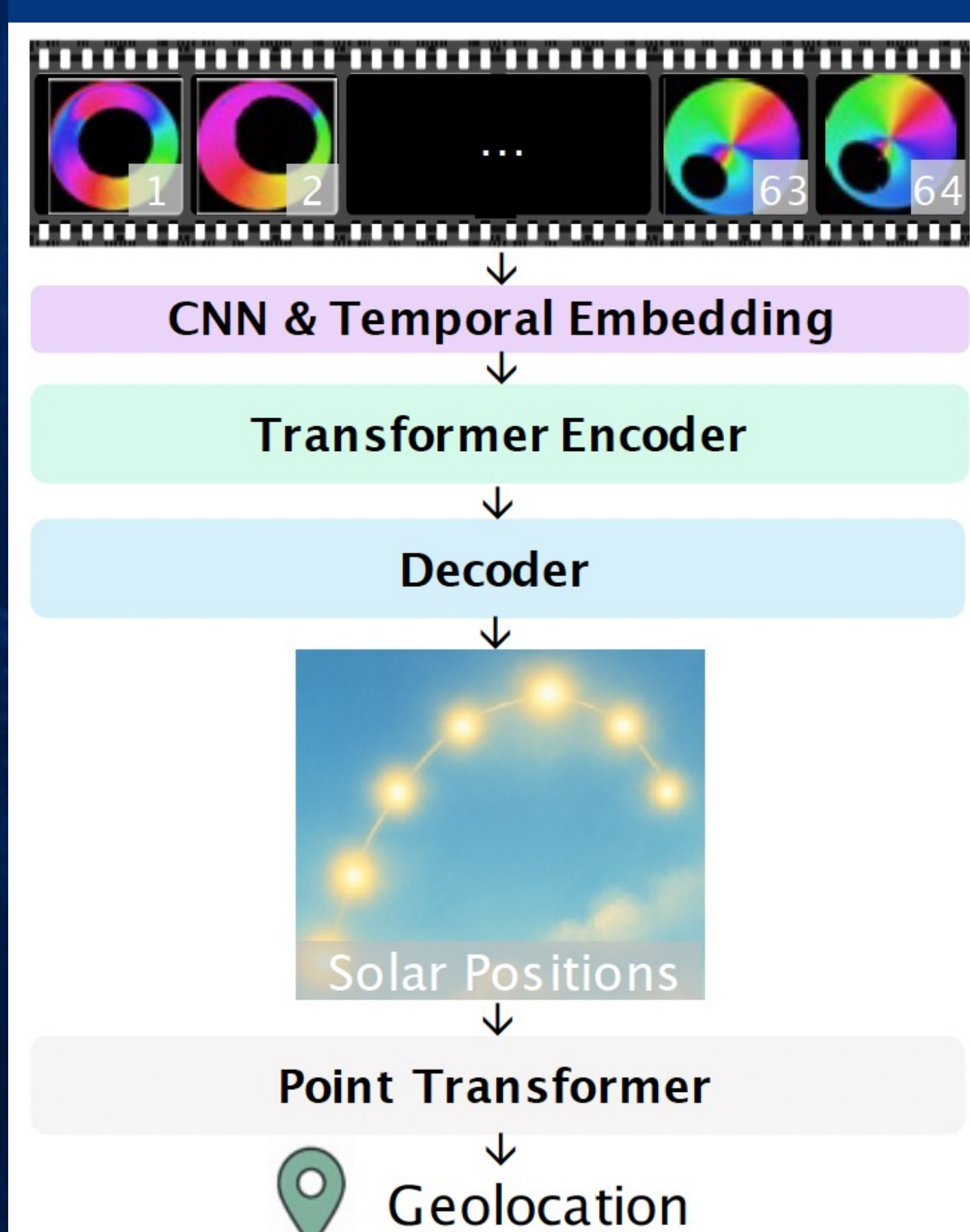
Collect data from a few optically diverse sites

Model Training Simulated Sites (January-December)

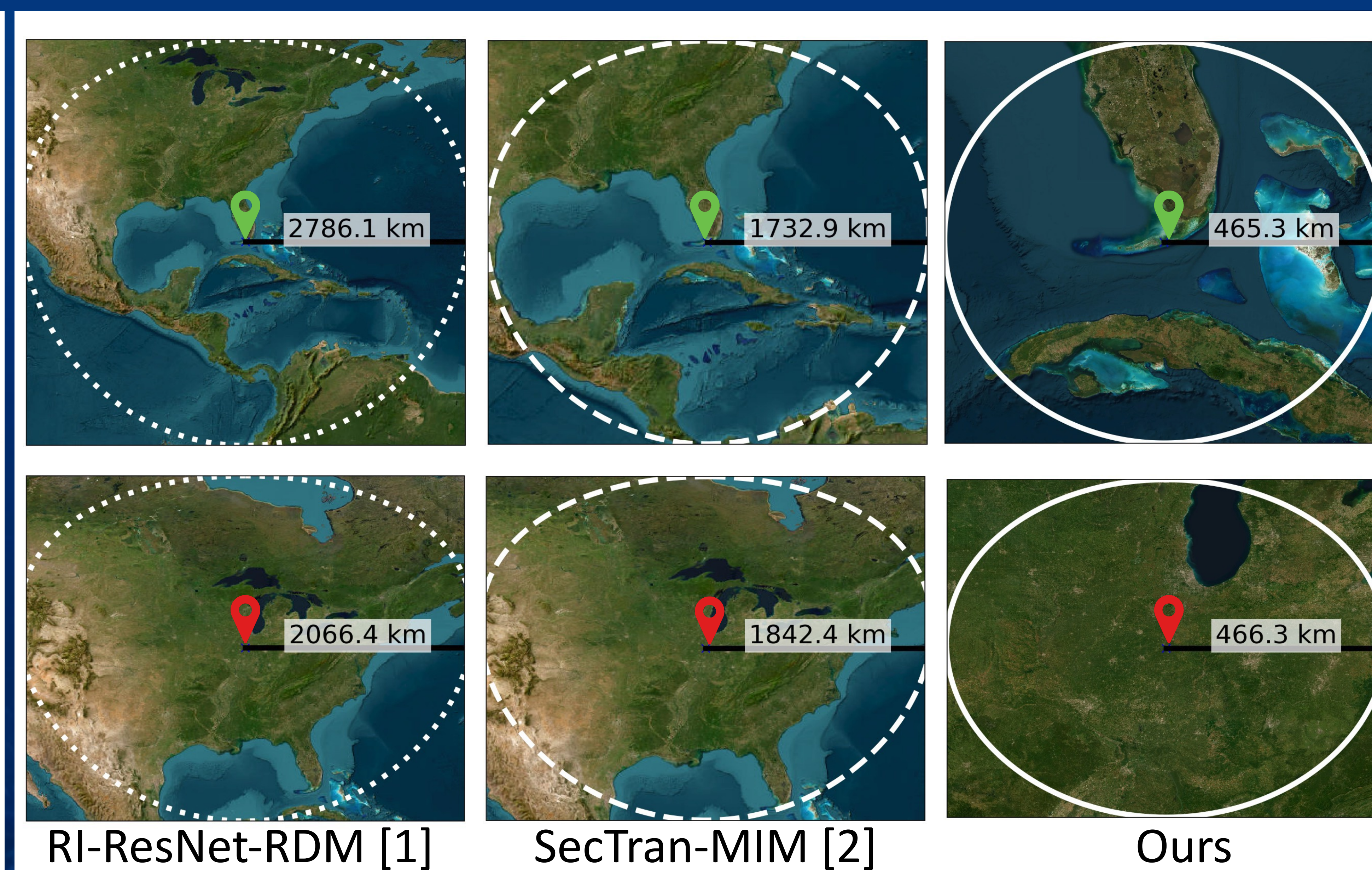
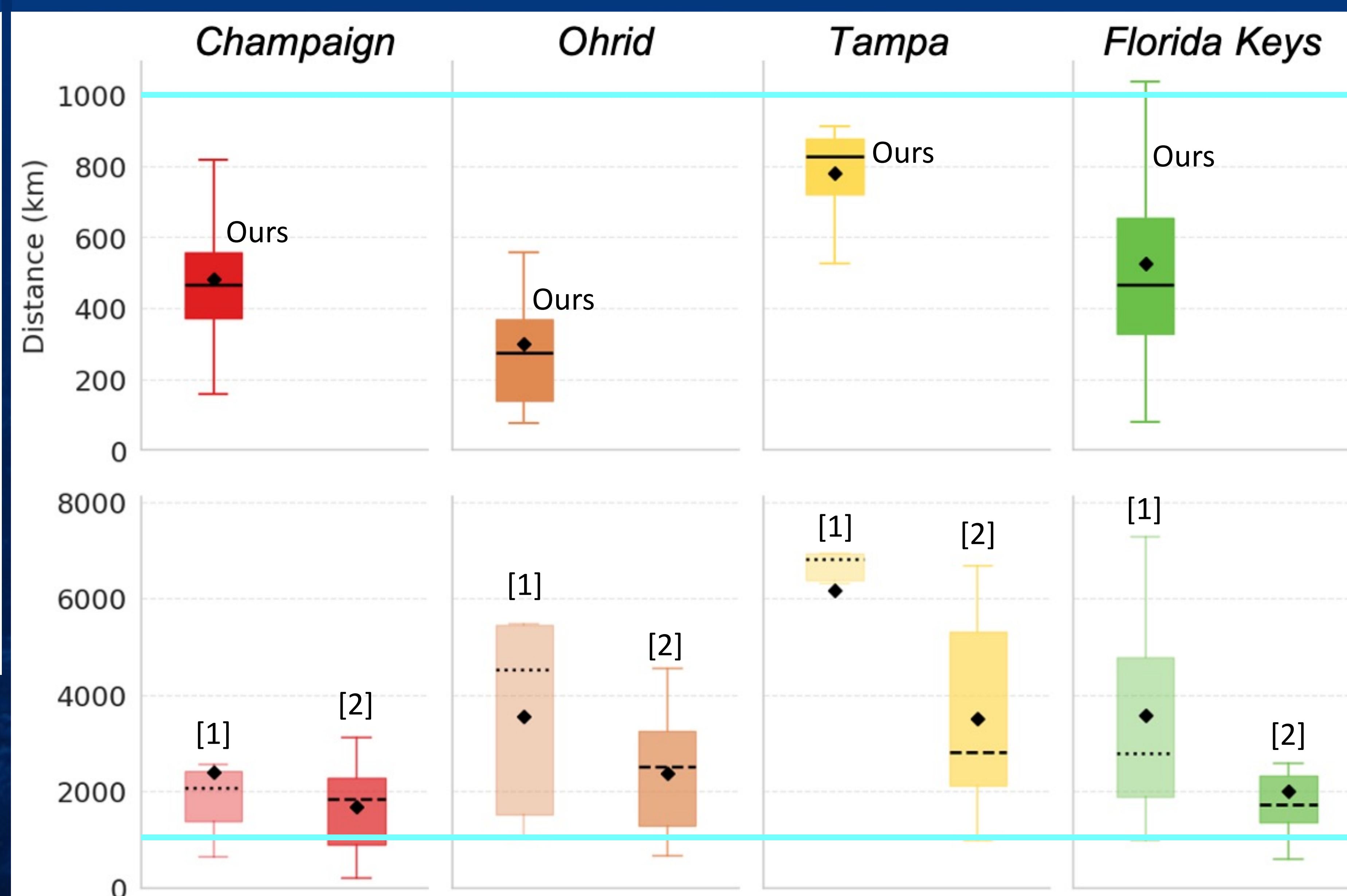


Assemble 64-frame *simulated training sequences* spanning latitudes, longitudes, seasons by matching solar elevations with varying waters

4. Polar Transformer



5. Results (Cross-Site Geolocation on Unseen Sites) Prior works generalize poorly



Additional details, more results and ablation study in the paper 😊